Prof. Hussein MA

THERAPEUTIC DRUG MONITORING (TDM)

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- is a clinical practice to keep a drug's concentration in the blood within a narrow window where it works best and is least toxic.

- It is used for drugs **with a narrow therapeutic range**, **high variability between patients**, or **serious toxicity risk**.

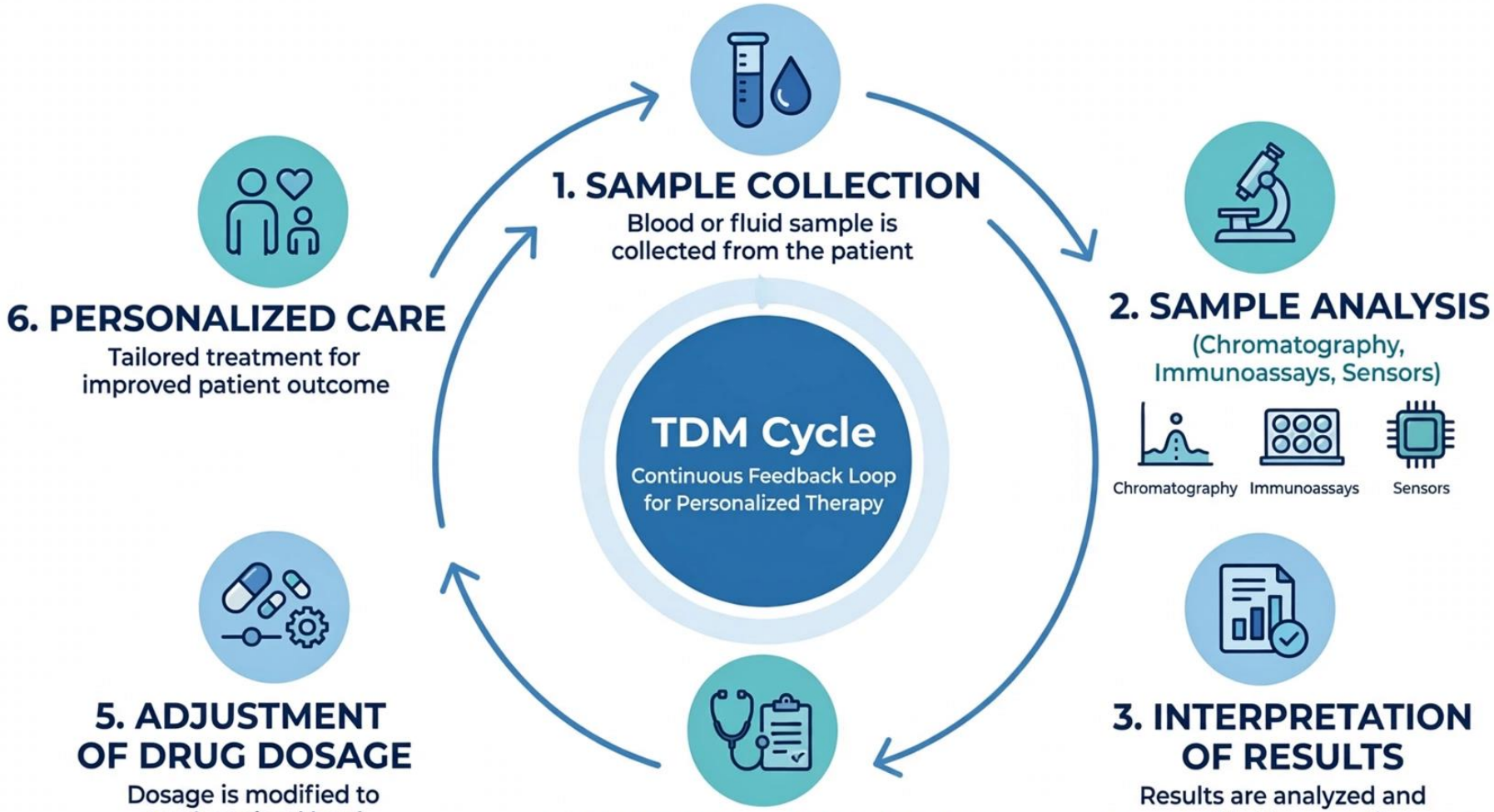
الهدف الرئيسي من TDM

المحافظة على تركيز الدواء داخل **المدى العلاجي** (Therapeutic Range)، بحيث:

- يكون التركيز كافيًا لإحداث التأثير العلاجي المطلوب.
- لا يكون مرتفعًا لدرجة تسبب أعراضًا سامة.

TDM Cycle

Therapeutic Drug Monitoring — Cyclical Process for Optimizing Medication



Goal: Maintain drug concentration within the therapeutic range
→ Maximize efficacy & minimize toxicity

1. Sample Collection

- A blood sample or other biological fluid is collected from the patient.
- The exact time of sampling relative to dose intake must be recorded precisely.
- For many drugs, sampling at the lowest concentration, before the next dose, is preferred.

2. Sample Analysis

The drug concentration is measured using laboratory techniques such as:
- Chromatography: e.g., HPLC and LC-MS/MS, the most accurate and specific
- Immunoassays: fast and automated
- Biosensors: new rapid point-of-care technologies

3. Interpretation of Results

The result is compared with the drug's known therapeutic range.
It is determined whether the concentration is:

- **Subtherapeutic**: below required, weak efficacy
- **Within therapeutic range**: desired target
- **Supratherapeutic / Toxic**: above range and may cause toxicity

4. Medical Judgement (Clinical Evaluation)

Decisions are not based on lab results alone. The physician considers:

- **Age and weight of the patient**
- **Liver and kidney functions**
- **Coexisting diseases**
- **Other concomitant medications and drug interactions**
- **Clinical response to treatment and signs of toxicity**
- **Patient adherence to dosage**

5. Adjustment of Drug Dosage

Based on results and clinical evaluation:

- **Increase the dose if concentration is low**
- **Decrease the dose if concentration is high**
- **Change the dosing interval**
- Temporarily hold the drug in case of toxicity
- Continue the same dose if the concentration is appropriate

THRIVING IN UNCERTAINTY

Adapting to change

Businesses that embrace agility respond more quickly and effectively to market shifts, customer demands, and unforeseen challenges.

Real-world impact

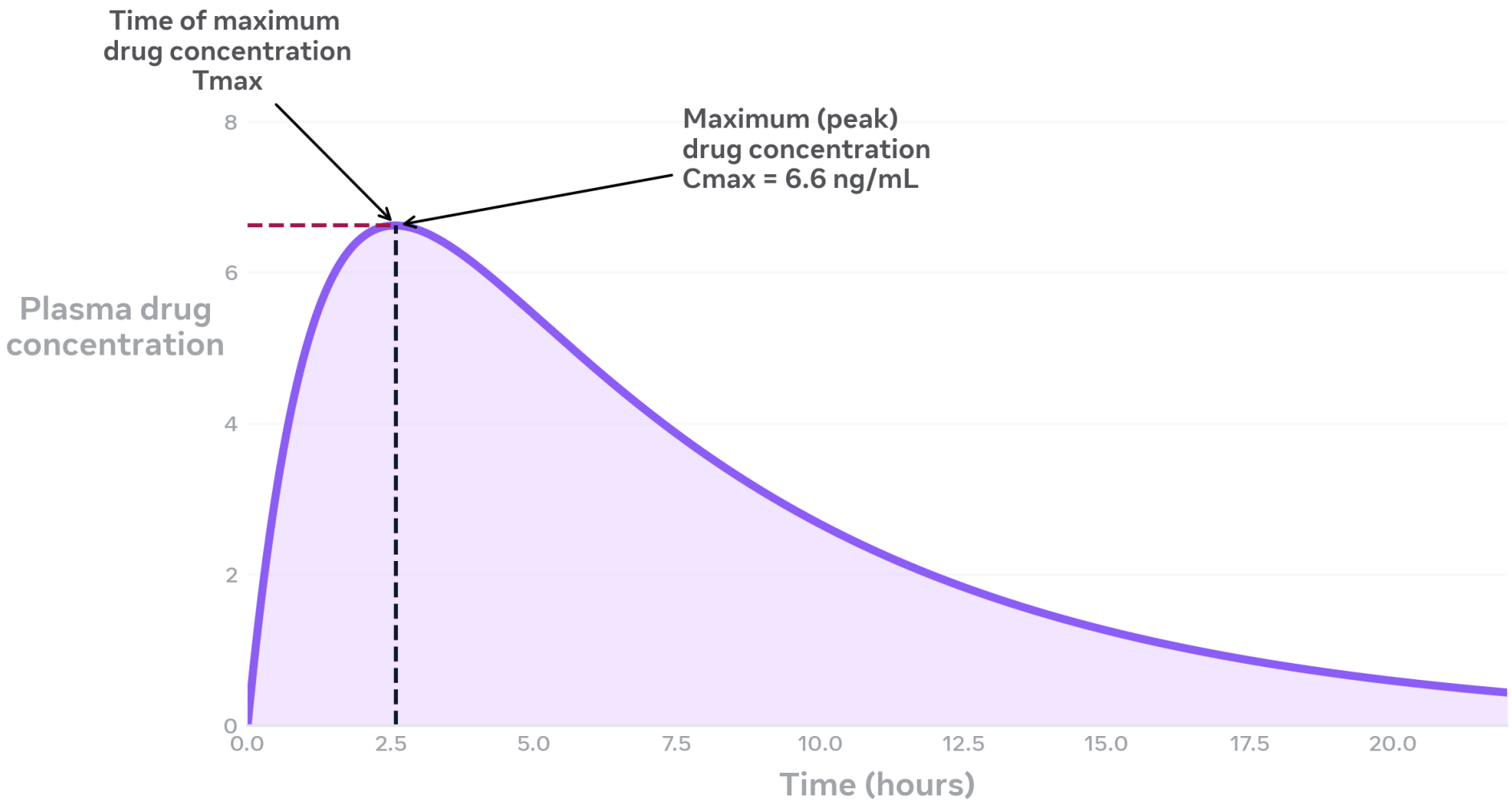
Industry leaders who stay ahead of disruption are those who proactively embrace change and continuously innovate.



Examples of drugs that require TDM

- Vancomycin
- Gentamicin and other aminoglycosides
- Digoxin
- Theophylline
- Carbamazepine
- Sodium valproate
- Phenytoin
- Tacrolimus and Cyclosporine after organ transplantation
- Lithium

Plasma Drug Concentration vs. Time Curve (Pharmacokinetic Profile)



Drug Sources

Source	Easy Explanation	Examples / Key Point
1. Plant Sources	Drugs from plant parts: seeds, stem, roots, leaves, resin	Digoxin from Digitalis (heart), Morphine from Opium (pain)
2. Animal Sources	Drugs taken from animals	Insulin (diabetes), Thyroid hormone
3. From Micro-organisms	Drugs made by molds & bacteria	**Penicillin**
4. Mineral Sources	Drugs from minerals / salts	**Lithium carbonate** (antipsychotic), MgSO4 (laxative)
5. Synthetic Sources	Man-made in lab. Copied from nature, made stronger with less side effects	**Sulfonamides, Aspirin**
6. Recombinant Proteins	Modern biotech: gene put inside cells to make protein drugs	**Interferons, Antibodies**

Drug Nomenclature

Type of Name	Easy Meaning	Example
Chemical Name	The full scientific description of the drug's chemical structure. Very long and	N-acetyl-para-aminophenol (APAP = NAPAP)
Generic Name	Shorter, simpler name. Made from the chemical name. Official name used w	Paracetamol = Acetaminophen (same drug, 2 generic names)
Brand Name (Trade Name)	Name given by the company that makes it. The company's exclusive proper	Panadol® is the brand name for APAP Company owns this name

Drug Dosage Forms



Tablets





Aerosol

Capsule



Suspension

Injection



Cream

Infusion



Solution





Basic Pharmacology Definitions

Term	Easy Meaning	Simple Example
Mechanism of Action	How the drug works to give its effect in the body	Paracetamol blocks pain signals in the brain
Dose	How much drug you give at ONE time	500 mg tablet x 1
Duration of Action	How long the drug stays working	Effect lasts for 6 hours
Onset of Action	How fast the drug starts working after you take it	Starts working after 30 minutes
Medication	Any substance given to help a patient	Used for diagnosis, cure, treatment, or prevention
Prescription	The written paper from the doctor that tells the pharmacist how to prepare and give the drug	The "روشتة" - Doctor's order

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Classes of Drugs

Class	Can you buy it without doctor?	Easy Meaning	Example
1. Over The Counter (OTC)	YES, CAN buy without prescription	You can go to pharmacy and buy it directly. No need for doctor's paper. Safe for self-use.	Panadol, Vitamins, Cough syrup, Antacids
2. Prescription Only Medication (POM) روشتة	NO, CANNOT buy without prescription	You must have a written note (روشتة) from your doctor. Pharmacist will not give it to you without it.	Antibiotics, Blood pressure drugs, Insulin

Therapeutic Effect vs Side Effect

Type	Easy Meaning	Simple Definition
Therapeutic Effect	**The GOOD effect** - Why we take the drug. **The primary**, INTENDED effect. The reason the doctor prescribed it.	**Example**: We take Morphine sulfate for analgesia (pain relief). Pain relief is the therapeutic effect.
Side Effect	**The OTHER effect** - Not intended. **Secondary**, UNINTENDED effect. Usually predictable. Can be harmful or harmless.	**Example**: The same drug may cause other things you didn't want.

Type of Action		Easy Explanation	Example from slide
Therapeutic Effect	1. Stimulating or inhibiting cell function or blocking biochemicals	Drug stimulates or stops cells from working	Antihistamine - blocks histamine
	2. Physical or mechanical action	Drug works by physical movement, not chemical	Laxative - pushes stool out
Side Effect	1. Dangerous / tissue damage / life-threatening / toxic	Very serious side effects, can harm the body	Excessive bleeding - can be fatal
	2. Mild side effects	Small, bearable side effects, not dangerous	Antihistamines may cause dry mouth

Indications vs Contra-indications

Term	Easy Meaning	In Simple Words	Example
Indications	WHEN TO USE the drug	"The reasons for giving a medication or doing a treatment" - Why you SHOULD use it	You have bacterial infection → Indication for antibiotic is infection
Contra-indications	WHEN NOT TO USE the drug	"Factor that prevents the use of a medication or treatment" - Why you should NOT use it	You are allergic to penicillin → Allergy is Contra-indication for penicillin